

Instructor: Date:

Learner: Regional Branch:

Program: Center:

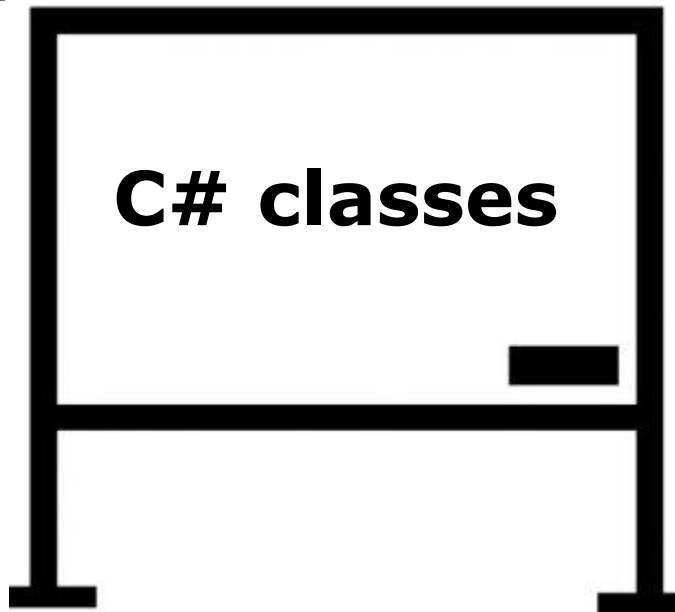
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Let's Start!

Functions in C#



Learning Outcome:

Talk about principles, concepts, characteristics of general-purpose computer programming language in spoken and written situations.

Ask your partner:

- What keywords do you know for C#?
What are they for?
- What are member variables?

Can Do Statements:

- I can define a class in C#.
- I can talk about member functions and encapsulation in C#.
- I can talk about constructors and destructors in C#.





Let's Explore!

Lisa talks about her next class on C#.



Instructions: Read the text below.

- Johnson:** Hello, Lisa. Congratulations on your course! Everyone loved it!
- Lisa:** Thank you, Sir. I'm very pleased, although I have to admit I was a bundle of nerves before the course took place.
- Johnson:** Let's talk about your next course. What are you going to explain?
- Lisa:** I believe the students need to learn the concept of "classes." They need to know how to define a blueprint for a data type. By defining this, they will know what an object of the class consists of and the operations that can be performed on that object.
- Johnson:** I see! What are you going to say about syntax?
- Lisa:** I am going to write and explain the syntax. I will write the keyword plus the class name and then the class body enclosed by curly braces. Access specifies the access rules for the members and the class itself. When not mentioned, the default access specifier for a class type is internal.
- Johnson:** Great skills, Lisa! What else are you going to include in the presentation?
- Lisa:** Class constructor and destructor.
- Johnson:** Why are you going to include these two topics?
- Lisa:** Because a class constructor has exactly the same name as that of the class and doesn't have a return type. It is an essential function of a class that is executed whenever we create a new object of that class. On the other hand, a destructor is executed whenever an object of its class goes out of scope; it has the same name of the class and the prefixed tilde.
- Johnson:** Are you adding anything else?
- Lisa:** The other topic will be the static member. Do you want me to talk about anything in special?
- Johnson:** I think these three topics will be enough.
- Lisa:** Where is the presentation going to take place this time?
- Johnson:** I don't have that information yet.
- Lisa:** I heard I'm not going to be the only presenter. Who will be on the platform with me?
- Johnson:** I suppose it will be Alison. Be prepared; the presentation is on Tuesday morning.



**Let's Practice!**

Talking about classes in C#

Instructions: Select the correct answer.

1. People (has / have /having) to learn about classes.
2. They (needs / needing / need) to know how to define a blueprint for a data type.
3. It (are / be / is) important that they clarify what an object of the class consists of.
4. They also need to identify what operations (are / Is / being) performed on that object.
5. A class (let / letting / lets) you create your own custom types.

**Let's Practice!**

Clarifying constructors and destructors in C#!

Instructions: Read the sentences. Underline the correct option to complete the sentence.

1. (Why you go / Why do you go / Why are you going) to include class constructor and destructor in your presentation?
2. Who is (going to clarifying / goes to clarify / going to clarify) the syntax?
3. (What are / Which are / When are) the keywords going to be enclosed by curly braces?
4. What is going to (are within the keyword/be within the keyword / is within the keyword) and class name? the class body?
5. What is (going to specify / going to specifying / going to specifies) the access rules for the members? The access?



Extension Activity

Explaining How a Class Works

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search for some information on the web about how a class works.
- Make a poster by using different colors and few words explaining the definition and features of a class.
- Make a different poster illustrating the process of a class, how it works and how it is related to objects.
- Be ready to present your posters and explain your ideas in class.

Self-Assessment

Yes

No

1. I define a class in C#.

☐
☐

2. I talk about member functions and encapsulation in C#.

☐
☐

3. I explain constructors and destructors in C#.

☐
☐

4. I mention some feature of a class in C#.

☐
☐


Target Words

- **take place** (verb): to happen
e.g. Where is the presentation going to take place this time?
- **a bundle of nerves** (expression): to be very nervous
e.g. I'm very pleased although I have to admit I was a bundle of nerves before the course took place.
- **blueprint** (noun): an initial outline of a project
e.g. They need to know how to define a blueprint for a data type.
- **to go out of scope** (expression): for an object, to become not visible
e.g. On the other hand, a destructor is executed whenever an object of its class goes out of scope.
- **tilde** (noun): a keyword (~)
e.g. On the other hand, a destructor is executed whenever an object of its class goes out of scope; it has the same name of the class and the prefixed tilde.



Learning Points

The Future with Going to - Information Questions

The future with going to in interrogative form can also be used with information questions.

We can create questions in the future with **going to** to ask for specific information on someone or something.

To do this, we can follow one of these structures depending on the kind of information we are trying to get:

Wh word + verb to be + subject + **going to** + complement

Example: ***Who** is **going to** help me with the presentation?*

Wh word + object + verb to be + subject + **going to** + complement

Example: ***What** class name are you **going to** use to start that object?*

In the table below, you will find every **wh word** you can use to ask for information with the explained structure.

What	What are you going to present?
Which	Which class are you going to work on?
Where	Where are you going to run the code?
Why	Why is she going to use that computer?
How	How are they going to work on the new code?
Who	Who is going to help me with the code?
When	When are we going to finish this new task?

Examples

- *What are you going to explain?*
- *Where is the presentation going to take place this time?*
- *Why are you going to include these two topics?*

Static Members of a C# Class

We define class as static using the **static** keyword.

The keyword **static** implies there is only one copy of the static member. Static variables are used for defining constants because their values can be taken out by invoking the class without creating an instance of it. It can be initialized outside the member function and outside or inside the class definition.

